

VOLUME CONSTANTS FOR STEPPING-STONE EMPTY REGIONS IN $\mathbb{R}^d$

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ABSTRACT. We prove an explicit normalized one-dimensional integral formula for the d -dimensional volume of the stepping-stone empty region

$$S_{\text{SS},\alpha}(p, q) = \left\{ z \in \mathbb{R}^d : \|z - p\|^\alpha + \|z - q\|^\alpha \leq \ell_{p,q}^\alpha \right\},$$

where $d \geq 2$ is the ambient dimension, $\ell_{p,q} = \|p - q\|$, and $\alpha \geq 1$ is the stepping-stone exponent. The final result is

$$a_{d,\text{SS}}(\alpha) = 2\kappa_{d-1} \int_{2^{-1/\alpha}}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du.$$

1. NOTATION AND PROBLEM

Let $d \geq 2$ be the ambient Euclidean dimension and let $\alpha \geq 1$ be a real parameter. We write λ_d for d -dimensional Lebesgue measure on $\mathbb{R}^d$ and

$$\kappa_m := \lambda_m(B_m(0, 1)) = \frac{\pi^{m/2}}{\Gamma(m/2 + 1)}$$

for the volume of the unit ball in $\mathbb{R}^m$.

Definition 1 (Stepping-stone region in $\mathbb{R}^d$). *For distinct $p, q \in \mathbb{R}^d$, set $\ell_{p,q} := \|p - q\| > 0$. The stepping-stone region with exponent $\alpha \geq 1$ is*

$$(1) \quad S_{\text{SS},\alpha}(p, q) := \left\{ z \in \mathbb{R}^d : \|z - p\|^\alpha + \|z - q\|^\alpha \leq \ell_{p,q}^\alpha \right\}.$$

The normalized unit representative is

$$K_{\text{SS},\alpha} := S_{\text{SS},\alpha}(0, e_1) = \left\{ z \in \mathbb{R}^d : \|z\|^\alpha + \|z - e_1\|^\alpha \leq 1 \right\}.$$

The goal is to compute the unit-volume constant

$$a_{d,\text{SS}}(\alpha) := \lambda_d(K_{\text{SS},\alpha}),$$

so that

$$\lambda_d(S_{\text{SS},\alpha}(p, q)) = \ell_{p,q}^d a_{d,\text{SS}}(\alpha).$$

2. NORMALIZED COORDINATES

To compute $a_{d,\text{SS}}(\alpha)$ for the normalized configuration $p = 0, q = e_1$. Write a point $z \in \mathbb{R}^d$ as

$$z = (X, Z), \quad X \in \mathbb{R}, \quad Z \in \mathbb{R}^{d-1},$$

and define $Y := \|Z\| \geq 0$. Then

$$\|z\|^2 = X^2 + Y^2, \quad \|z - e_1\|^2 = (X - 1)^2 + Y^2.$$

The boundary is described by

$$(2) \quad (X^2 + Y^2)^{\alpha/2} + ((X - 1)^2 + Y^2)^{\alpha/2} = 1.$$

Instead of solving (2) directly for Y as a function of X , we parametrize the boundary by the distance to the first site. Let

$$u := \|z\| = \sqrt{X^2 + Y^2}.$$

On the boundary, the second distance is

$$\rho(u) := \|z - e_1\| = (1 - u^\alpha)^{1/\alpha}.$$

The right half of the body corresponds to

$$u \in [2^{-1/\alpha}, 1].$$

The lower endpoint is where the two distances are equal:

$$u = \rho(u), \quad u^\alpha + u^\alpha = 1, \quad u = 2^{-1/\alpha}.$$

3. BOUNDARY PARAMETRIZATION

The point $z = (X, Z)$ lies simultaneously on two spheres:

$$(3) \quad X^2 + Y^2 = u^2,$$

$$(4) \quad (X - 1)^2 + Y^2 = \rho(u)^2 = (1 - u^\alpha)^{2/\alpha}.$$

Subtracting (4) from (3) gives

$$2X - 1 = u^2 - \rho(u)^2.$$

Hence

$$(5) \quad X_\alpha(u) = \frac{1 + u^2 - (1 - u^\alpha)^{2/\alpha}}{2}.$$

It is convenient to shift the origin to the midpoint $1/2$ along the first coordinate:

$$(6) \quad x_\alpha(u) := X_\alpha(u) - \frac{1}{2} = \frac{u^2 - (1 - u^\alpha)^{2/\alpha}}{2}.$$

The transverse radius is obtained from (3):

$$(7) \quad y_\alpha(u) := Y(u) = \sqrt{u^2 - X_\alpha(u)^2} = \frac{1}{2} \sqrt{4u^2 - (1 + u^2 - (1 - u^\alpha)^{2/\alpha})^2}.$$

Differentiating (6),

$$(8) \quad x'_\alpha(u) = u + u^{\alpha-1}(1 - u^\alpha)^{(2-\alpha)/\alpha}.$$

Indeed,

$$\frac{d}{du}(1 - u^\alpha)^{2/\alpha} = \frac{2}{\alpha}(1 - u^\alpha)^{2/\alpha-1}(-\alpha u^{\alpha-1}) = -2u^{\alpha-1}(1 - u^\alpha)^{(2-\alpha)/\alpha}.$$

Thus

$$\frac{d}{du} \left[\frac{u^2 - (1 - u^\alpha)^{2/\alpha}}{2} \right] = u + u^{\alpha-1}(1 - u^\alpha)^{(2-\alpha)/\alpha}.$$

4. THE $\mathbb{R}^d$ INTEGRAL FORMULA

Theorem 1 (One-dimensional volume formula). *Let $d \geq 2$ and $\alpha \geq 1$. Then*

$$(9) \quad a_{d,SS}(\alpha) = 2\kappa_{d-1} \int_{2^{-1/\alpha}}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du,$$

where

$$(10) \quad y_\alpha(u) = \frac{1}{2} \sqrt{4u^2 - (1 + u^2 - (1 - u^\alpha)^{2/\alpha})^2},$$

and

$$(11) \quad x'_\alpha(u) = u + u^{\alpha-1}(1 - u^\alpha)^{(2-\alpha)/\alpha}.$$

Consequently, for arbitrary $p \neq q$,

$$(12) \quad \lambda_d(S_{SS,\alpha}(p, q)) = \ell_{p,q}^d \left[2\kappa_{d-1} \int_{2^{-1/\alpha}}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du \right].$$

Proof. Work in the normalized configuration $(p, q) = (0, e_1)$. The unit region $K_{\text{SS}, \alpha}$ is invariant under rotations fixing the e_1 -axis. Therefore each slice perpendicular to the e_1 -axis is a $(d-1)$ -dimensional ball.

Use midpoint-centered longitudinal coordinate $x = X - 1/2$. The right half of the body has $x \geq 0$ and is parametrized by $u \in [2^{-1/\alpha}, 1]$ via $x = x_\alpha(u)$. At each parameter value u , the transverse radius is $y_\alpha(u)$. Hence the $(d-1)$ -dimensional cross-sectional volume is

$$\kappa_{d-1} y_\alpha(u)^{d-1}.$$

The infinitesimal longitudinal thickness is

$$dx = x'_\alpha(u) du.$$

Therefore the right-half volume is

$$\int_{2^{-1/\alpha}}^1 \kappa_{d-1} y_\alpha(u)^{d-1} x'_\alpha(u) du.$$

The unit region is symmetric under reflection across the midpoint hyperplane $X = 1/2$, so the total normalized volume is twice the right-half volume:

$$a_{d,\text{SS}}(\alpha) = 2\kappa_{d-1} \int_{2^{-1/\alpha}}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du.$$

□

Corollary 1 (Recovery of the planar formula). *For $d = 2$, $\kappa_{d-1} = \kappa_1 = 2$, so (9) becomes*

$$a_{2,\text{SS}}(\alpha) = 4 \int_{2^{-1/\alpha}}^1 y_\alpha(u) x'_\alpha(u) du.$$

This is exactly the planar stepping-stone area formula.

5. COMPUTATIONAL EVALUATION OF THE INTEGRAL

The formula in [Theorem 1](#) is directly suitable for numerical quadrature. For fixed $d \geq 2$ and $\alpha \geq 1$, define

$$u_0 := 2^{-1/\alpha},$$

and evaluate

$$a_{d,\text{SS}}(\alpha) = 2\kappa_{d-1} \int_{u_0}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du.$$

The only ingredients are

$$y_\alpha(u) = \frac{1}{2} \left[4u^2 - (1 + u^2 - (1 - u^\alpha)^{2/\alpha})^2 \right]^{1/2}$$

and

$$x'_\alpha(u) = u + u^{\alpha-1} (1 - u^\alpha)^{(2-\alpha)/\alpha}.$$

Thus a numerical implementation needs only a one-dimensional adaptive quadrature routine. In floating-point arithmetic, the expression inside the square root for $y_\alpha(u)$ may become slightly negative near an endpoint because of roundoff; replacing it by its positive part does not change the mathematical integral.

One possible Python-style pseudocode is given in [Section 5](#). For example, in `scipy`, the placeholder `adaptive_quad` can be replaced by `scipy.integrate.quad`. The endpoint $\alpha = 1$ is degenerate and may be treated separately by returning $a_{d,\text{SS}}(1) = 0$.

Python-style pseudocode

```

1 def kappa(m):
2     return pi**(m / 2) / gamma(m / 2 + 1)
3
4 def y_alpha(u, alpha):
5     r = (1 - u**alpha)**(2 / alpha)
6     inside = 4*u**2 - (1 + u**2 - r)**2
7     return 0.5 * sqrt(max(inside, 0))

```

```

8
9 def x_alpha_prime(u, alpha):
10     return u + u**(alpha - 1) * (1 - u**alpha)**((2 - alpha) / alpha)
11
12 def a_ss(d, alpha):
13     if alpha == 1:
14         return 0
15
16     u0 = 2**(-1 / alpha)
17
18     def integrand(u):
19         y = y_alpha(u, alpha)
20         xp = x_alpha_prime(u, alpha)
21         return y**(d - 1) * xp
22
23     value = adaptive_quad(integrand, u0, 1)
24     return 2 * kappa(d - 1) * value

```

For example, in `scipy`, the placeholder `adaptive_quad` can be replaced by `scipy.integrate.quad`. The endpoint $\alpha = 1$ is degenerate and may be treated separately by returning $a_{d,SS}(1) = 0$.

6. ANCHOR VALUES

Proposition 1 (Degenerate endpoint $\alpha = 1$). *For every $d \geq 2$,*

$$a_{d,SS}(1) = 0.$$

Proof. When $\alpha = 1$,

$$\|z - p\| + \|z - q\| \leq \ell_{p,q}.$$

The reverse inequality is always true by the triangle inequality. Hence equality must hold, which occurs exactly on the segment $[p, q]$. A line segment has zero d -dimensional Lebesgue measure for $d \geq 2$. $\square$

Proposition 2 (Quadratic endpoint $\alpha = 2$). *For every $d \geq 2$,*

$$a_{d,SS}(2) = \frac{\kappa_d}{2^d}.$$

Proof. Let $m = (p + q)/2$. The parallelogram identity gives

$$\|z - p\|^2 + \|z - q\|^2 = 2\|z - m\|^2 + \frac{\ell_{p,q}^2}{2}.$$

Thus

$$\|z - p\|^2 + \|z - q\|^2 \leq \ell_{p,q}^2 \iff \|z - m\|^2 \leq \frac{\ell_{p,q}^2}{4}.$$

Therefore $S_{SS,2}(p, q)$ is the d -ball centered at m with radius $\ell_{p,q}/2$, and

$$\lambda_d(S_{SS,2}(p, q)) = \kappa_d \left(\frac{\ell_{p,q}}{2} \right)^d.$$

Dividing by $\ell_{p,q}^d$ gives $a_{d,SS}(2) = \kappa_d/2^d$. $\square$

Proposition 3 (Limit $\alpha \rightarrow \infty$). *As $\alpha \rightarrow \infty$,*

$$S_{SS,\alpha}(p, q) \uparrow B(p, \ell_{p,q}) \cap B(q, \ell_{p,q}),$$

and

$$a_{d,SS}(\infty) = \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right),$$

where $I_z(a, b)$ is the regularized incomplete beta function.

Proof. For nonnegative a, b ,

$$\lim_{\alpha \rightarrow \infty} (a^\alpha + b^\alpha)^{1/\alpha} = \max\{a, b\}.$$

Hence the limiting constraint is

$$\max\{\|z - p\|, \|z - q\|\} \leq \ell_{p,q},$$

which is equivalent to $z \in B(p, \ell_{p,q}) \cap B(q, \ell_{p,q})$.

For the normalized configuration, this is the intersection of two unit d -balls whose centers are separated by distance 1. The radical hyperplane is halfway between centers, so each ball contributes a spherical cap of height $1/2$. The volume of one cap of a unit d -ball with cap height h is

$$\frac{1}{2} \kappa_d I_{1-(1-h)^2} \left(\frac{d+1}{2}, \frac{1}{2} \right).$$

With $h = 1/2$, the beta argument is

$$1 - (1 - 1/2)^2 = 1 - 1/4 = 3/4.$$

There are two congruent caps, so

$$a_{d,SS}(\infty) = \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right).$$

□

7. MONOTONICITY AND BOUNDS

Proposition 4 (Monotonicity in α). *If $1 \leq \alpha_1 < \alpha_2$, then*

$$S_{SS,\alpha_1}(p, q) \subseteq S_{SS,\alpha_2}(p, q),$$

and therefore

$$a_{d,SS}(\alpha_1) \leq a_{d,SS}(\alpha_2).$$

Proof. For $a, b \geq 0$, the ℓ^r norm

$$(a^r + b^r)^{1/r}$$

is nonincreasing in r . If $z \in S_{SS,\alpha_1}(p, q)$, then

$$(\|z - p\|^{\alpha_1} + \|z - q\|^{\alpha_1})^{1/\alpha_1} \leq \ell_{p,q}.$$

By monotonicity of finite-dimensional ℓ^r norms,

$$(\|z - p\|^{\alpha_2} + \|z - q\|^{\alpha_2})^{1/\alpha_2} \leq (\|z - p\|^{\alpha_1} + \|z - q\|^{\alpha_1})^{1/\alpha_1} \leq \ell_{p,q}.$$

Thus $z \in S_{SS,\alpha_2}(p, q)$. Taking d -dimensional Lebesgue measure and using the factorization gives the monotonicity of $a_{d,SS}$. □

Corollary 2 (Universal bounds). *For all $d \geq 2$ and $\alpha \geq 1$,*

$$0 \leq a_{d,SS}(\alpha) \leq \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right).$$

Moreover, for $\alpha \geq 2$,

$$\frac{\kappa_d}{2^d} \leq a_{d,SS}(\alpha) \leq \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right).$$

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